

STATUS:PUBLIC | YOUTUBE & GITHUB



TrustMyBike: Edge-AI for Urban Road Quality Estimation

System Requirements Overview

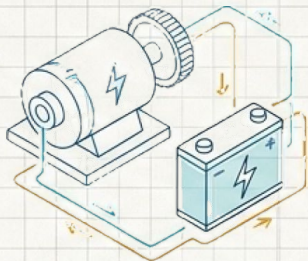


Energy Constraint

1

Total power < 6W

Must be self-sustaining via dynamo powering with reliable battery backup for stationary periods.

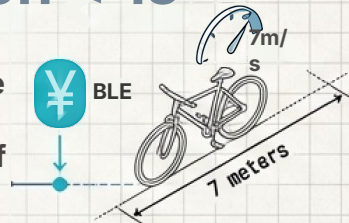


Latency Constraint

2

BLE Reconnection < 1s

Crucial tolerance: At an average 7m/s urban bike speed, losing $\geq 1s$ means missing >7 meters of road telemetry data.

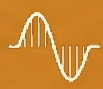
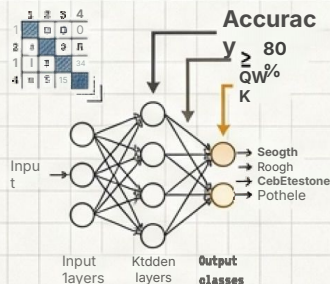


Functionality Target

3

**Accuracy $\geq 80\%$
AND QWK > 80%**

Requires strict adherence to minimum performance thresholds using Quadratic Weighted Kappa to account for ordinal road classes.

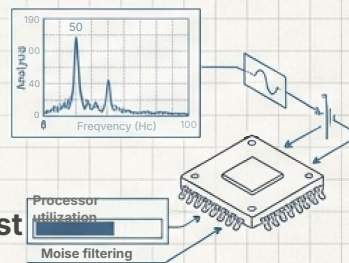


Sampling Frequency

4

Fixed at 50 Hz

Parameter established via FFT analysis to perfectly balance vibration noise reduction against embedded computational load.

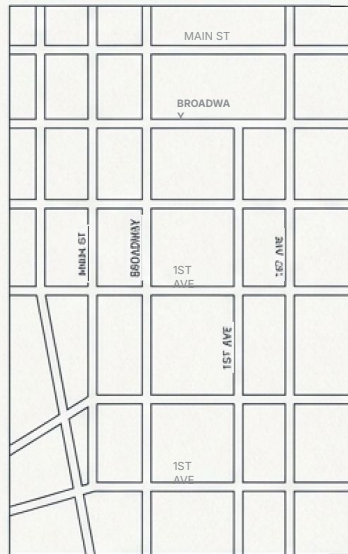


THE CORE FRICTION:MASSIVE DATA VS.MOBILE CONSTRAINTS

THE UNKNOWN ASPHALT

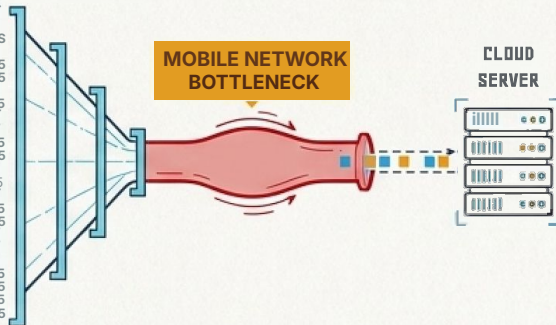


Cyclists and urban planners lack real-time, continuous data on physical road surface quality.



THE TECHNICAL BARRIER

100Hz 6-axis IMU raw data

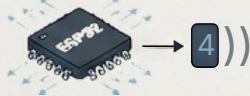
[illegible]

Continuous high-frequency vibration data is too heavy to stream via cellular networks. Sending raw data to the cloud drains battery, introduces massive latency, and guarantees data loss in urban dead zones.

TRUSTMYBIKE: FROM VIBRATION TO GLOBAL MAP

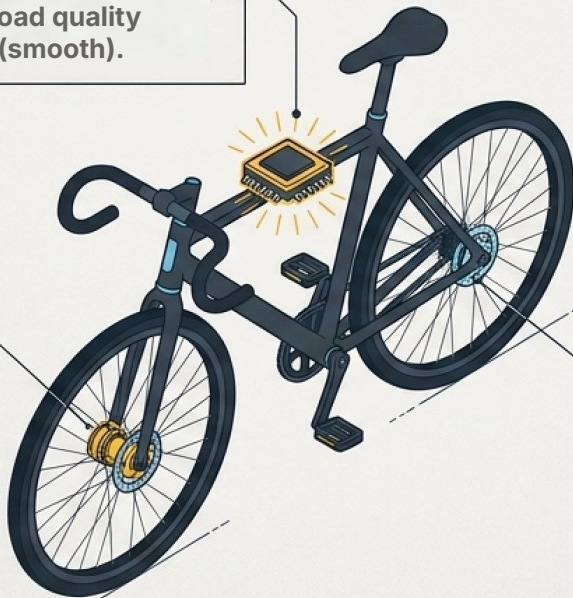
2. THINK

A local 1D-CNN processes the time-series window natively on-device, classifying road quality from 1 (terrible) to 5 (smooth).



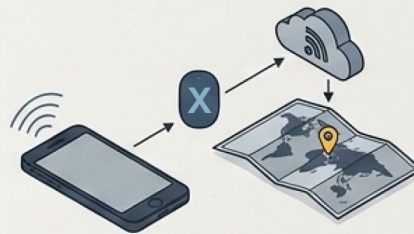
1. SENSE (THE HARDWARE)

The ESP32 microcontroller gathers localized IMU acceleration and velocity data directly from the wheel hub.

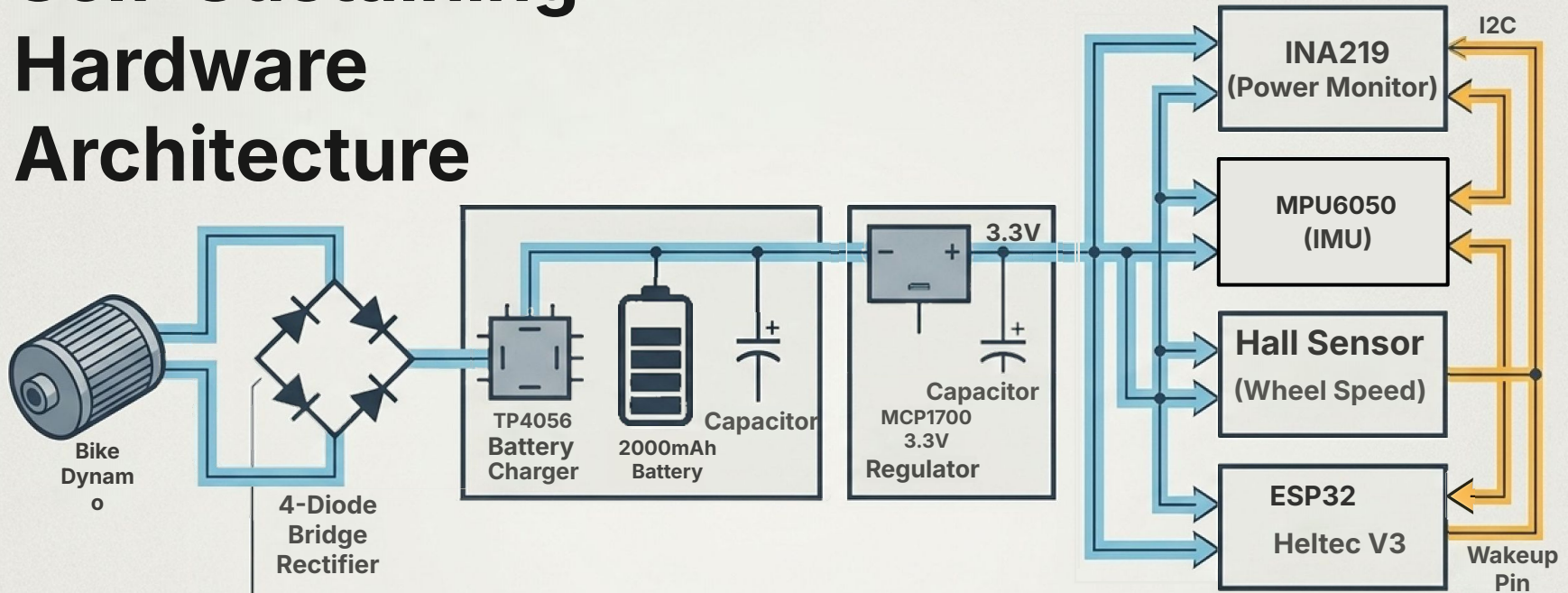


3. SHARE (THE NETWORK)

The system transmits only the lightweight classification integer via BLE to the rider's smartphone, appending GPS coordinates and forwarding it to the MQTT server.

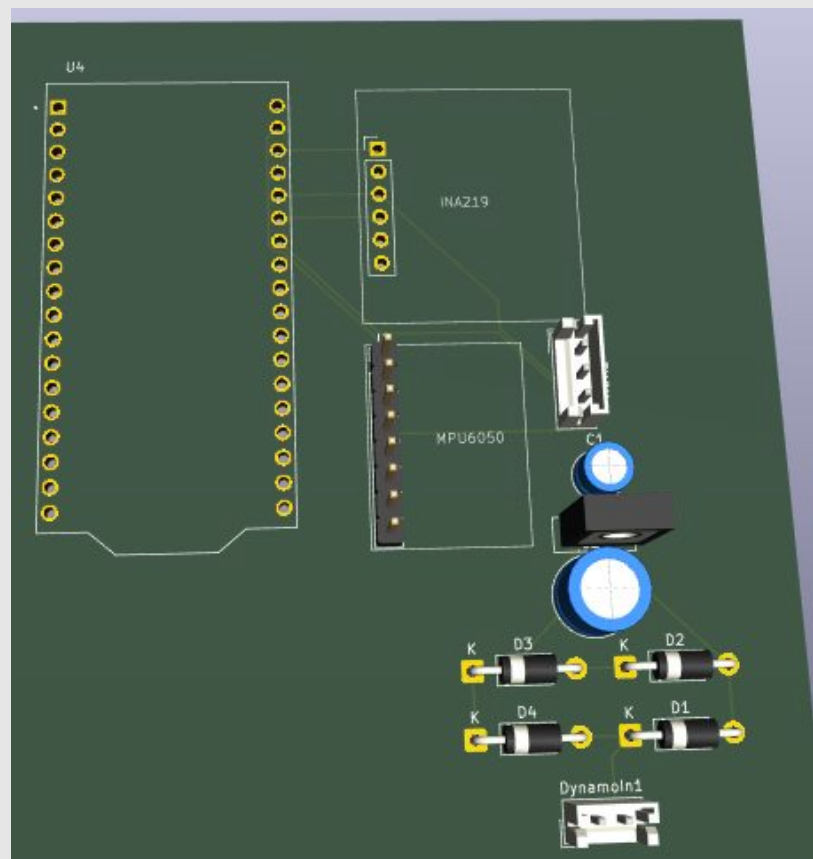
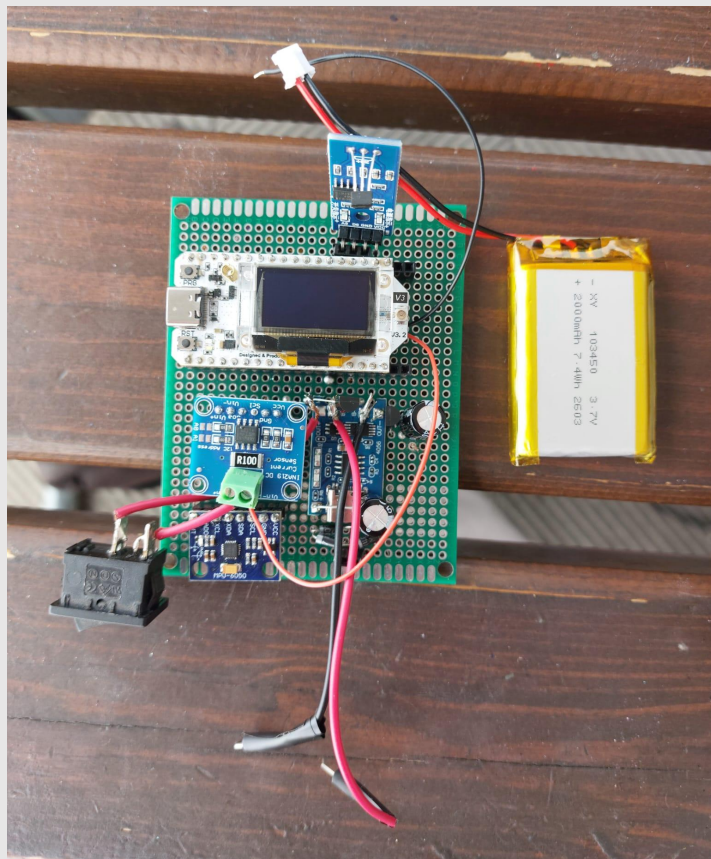


Self-Sustaining Hardware Architecture

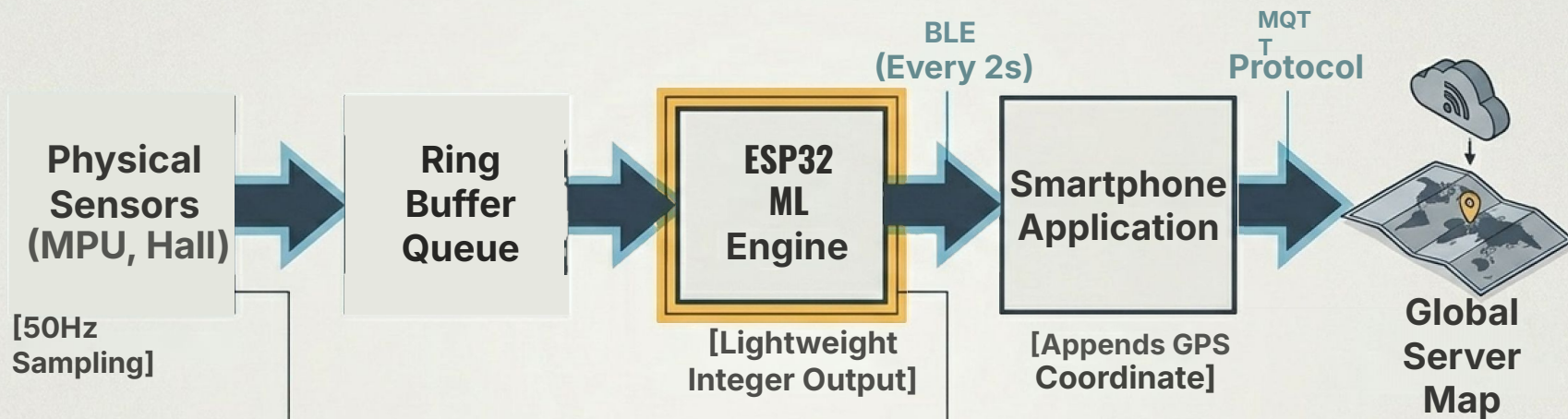


Power Generation: The dynamo feeds a 4-diode bridge rectifier to ensure the 2000mAh battery remains charged during transit.

Smart Sensing: ESP32 Heltec V3 acts as the master, polling the MPU6050 and INA219 via I2C. The Hall sensor measures velocity and functions as a hardware wakeup trigger.



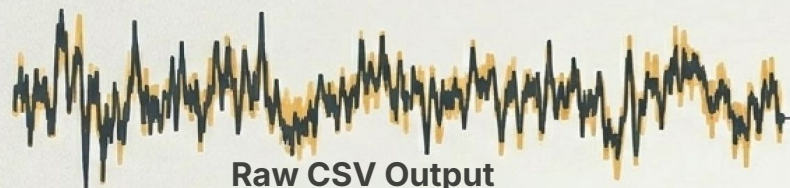
NETWORK & END-TO-END DATA FLOW



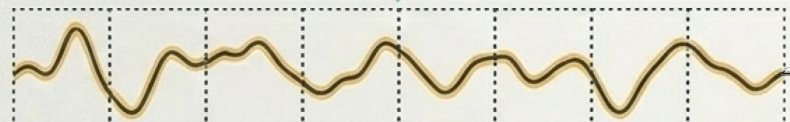
The Edge Advantage: Pushing raw 50Hz sensor data into a local Ring Buffer and executing the Machine Learning classification before transmission drastically collapses bandwidth requirements.

Asynchronous Reporting: Quality results are asynchronously broadcast via Bluetooth Low Energy (BLE) every 2 seconds. The phone assumes the heavy burden of GPS tagging and server synchronization.

Software Pipeline: Shaping the Signal



Raw CSV Output
(100Hz)



Downsampled (50Hz) & Windowed (64-192
samples)



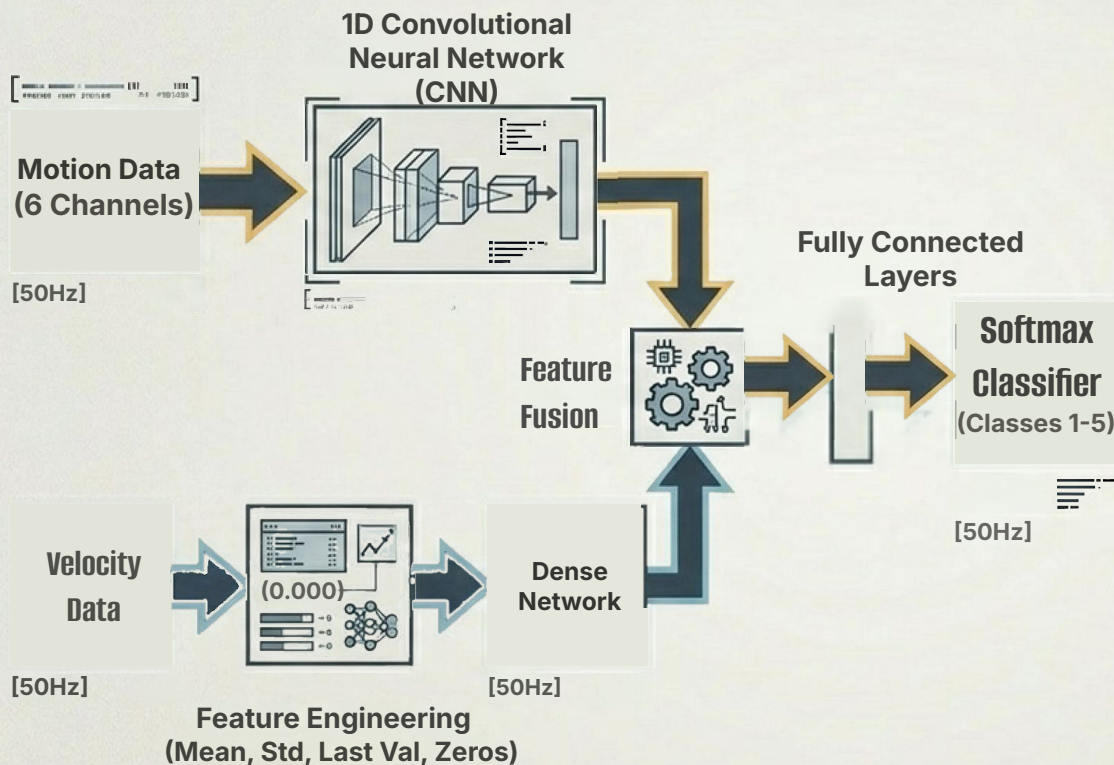
Grouped
Cross-Validation

Data Preparation: Raw CSV inputs (accelerometer, gyroscope, velocity) are cleaned and downsampled from 100Hz to 50Hz to suppress noise and reduce the embedded computational payload.

Time-Series Windowing: The continuous signal is partitioned into overlapping time windows. Road quality manifests as a temporal vibration pattern, necessitating windowed analysis rather than single-point evaluation.

Grouped Cross-Validation: Essential for model integrity. Signal windows extracted from the exact same recording never appear simultaneously in both training and validation sets, absolutely preventing data leakage.

Fusing Signal with Domain Knowledge



Branch A (Motion):

A 1D Convolutional Neural Network extracts complex temporal vibration patterns directly from the raw IMU waveform.

Branch B (Velocity):

A lightweight dense network processes engineered summary statistics derived from the vehicle's speed.

Why Dual-Branch?

High-amplitude vibrations signify different conditions at high speeds versus low speeds. Physically separating the inputs allows the architecture to contextually weigh vibration severity against forward momentum.

 **On-Device Execution**
TensorFlow Lite Micro is deployed to run inference entirely natively on the embedded ESP32 microcontroller.

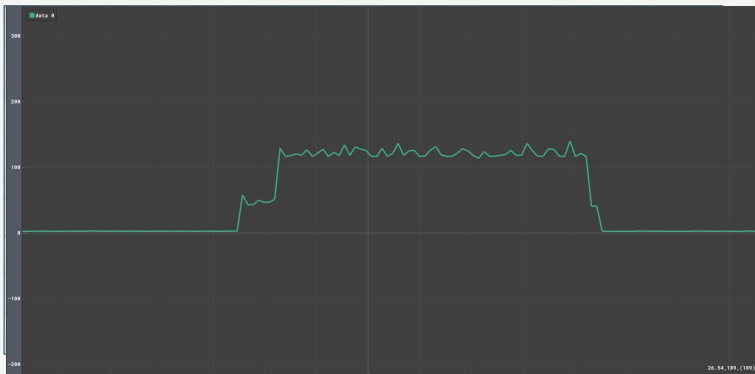
Power Optimization

The system evaluates a continuous score, applies temporal smoothing, and broadcasts via BLE. It triggers deep sleep instantly when the bicycle becomes stationary.

NoteboOk
M

System Evaluation: Power & Speed

Energy Profile Assessment



Target: $\leq 6W$.

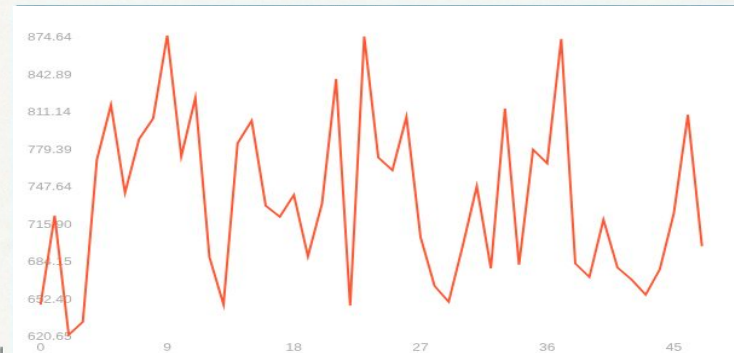


Result:

The system averages 369 mW active consumption (120mA * 3.3V) and 6.6 mW consumption in deep sleep. The onboard 2000mAh battery fully supports 10 hours of active tracking paired with 14 hours of deep sleep.



Network Latency Assessment



Target: $< 1s$ BLE reconnection.

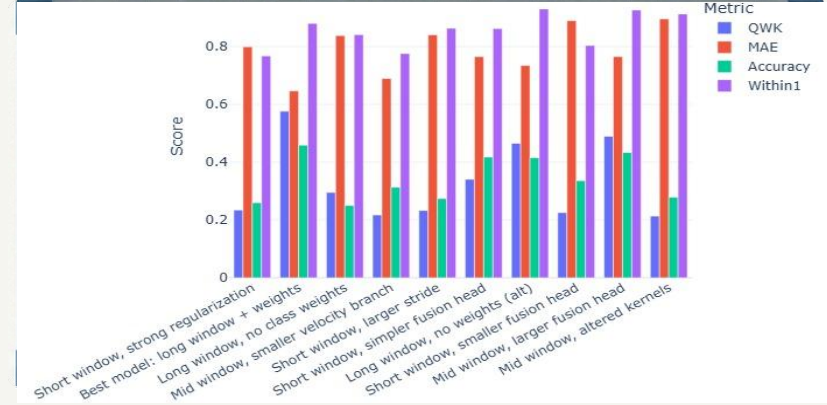


Result:

Accomplished.
At an average 7 m/s urban cycling speed, a 1-second delay correlates to roughly 7 meters of unrecorded road. This margin falls strictly within the standard error tolerance for commercial mobile GPS systems, rendering faster reconnection obsolete.



ML Evaluation: Accuracy & Quality Scoring



1

Accuracy
Target: $\geq 80\%$
Result: 50%

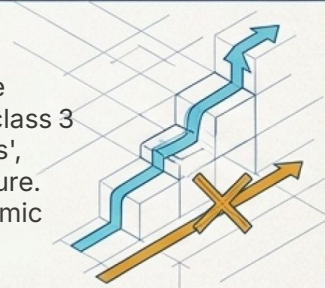
2

QWK
Target: $\geq 80\%$
Result: 70%

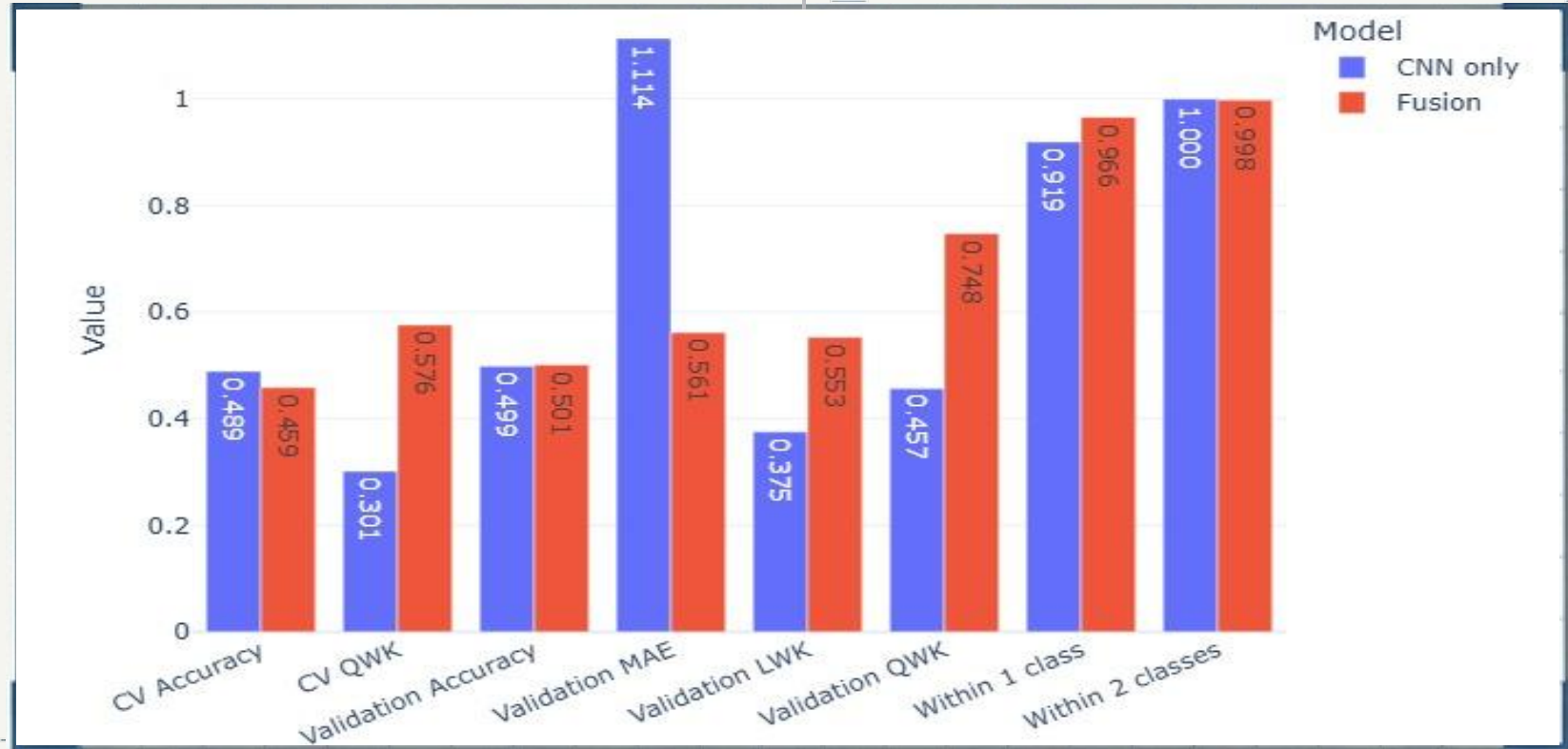
3

Interpreting the Matrix:

We employ Quadratic Weighted Kappa (QWK) because road surface quality is an ordinal metric. Predicting a class 3 on a class 4 road is evaluated as a tolerable 'near-miss', whereas predicting a 1 on a 4 is a critical systemic failure. The generated confusion matrix confirms that algorithmic errors remain tightly clustered around the true class.



ML Evaluation: Accuracy & Quality Scoring



50mm 20mm 30mm



-
- | Age group | Number of people |
|-----------|------------------|
| 0-14 | 100 |
| 15-24 | 150 |
| 25-34 | 200 |
| 35-44 | 250 |
| 45-54 | 300 |
| 55-64 | 350 |
| 65-74 | 300 |
| 75-84 | 200 |
| 85+ | 100 |

Expanding the Architecture: Future Scope

Sensor Fusion (Kalman Filter)

Moving beyond reliance on the basic Hall sensor to dynamically estimate velocity. Fusing the raw IMU data with wheel speed telemetry via a **Kalman filter** will vastly increase robustness against hardware sensor dropouts.

Dataset Generalization

Gathering a highly robust **multi-city** data sample. Expanding the geographic diversity of the training set to encompass varying global asphalt compositions will drastically improve the **generalized accuracy** of the core D-CNN.



TrustMyBike

Thank you for your time and attention.
The full project video is available publicly on YouTube.